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Neural Radiance Fields for Plot-level Cotton Crop Three-dimensional Reconstruction and Yield Estimation

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ABSTRACT. Cotton boll counting is a critical component of cotton yield prediction and plays a key role in advancing plant phenomics and precision agriculture. However, the complex distribution of cotton bolls and occlusions in 2D images pose significant challenges to fast and accurate counting. To overcome these challenges, a novel Neural Radiance Fields (NeRF)-based workflow was developed, integrating data acquisition, 3D reconstruction, and cotton boll mapping at the field-plot level. Two methods were employed for red-green-blue (RGB) image collection. The first used 10 cameras arranged in a single row and evenly mounted on a custom robotic platform to scan the cotton plants. The second utilized a smartphone for multi-angle scanning around plot-level cotton plants to capture complete plant videos. NeRF was then applied to reconstruct 3D models of plot-level cotton plants using the RGB images and export point clouds, and the reconstruction quality of the two data collection methods was compared. Next, an efficient data annotation method based on the Segment Anything Model (SAM) was designed to generate 2D masks of cotton bolls. Finally, cotton boll counting for each plot was achieved using improved FruitNeRF. In the experimental results, both data acquisition methods enabled effective 3D reconstruction. For the 3D reconstruction from smartphone data, the Peak Signal-to-Noise Ratio (PSNR) was 16.21, the Structural Similarity Index Measure (SSIM) was 0.34, and the Learned Perceptual Image Patch Similarity (LPIPS) was 0.49. The semi-automatic 2D mask annotation designed in this experiment reduced the labeling workload, and the final mean Average Precision at IoU 0.50 (mAP50) for cotton boll segmentation was 0.85. In yield estimation, the Mean Absolute Percentage Error (MAPE) between the predicted and ground truth number of cotton bolls was 4.41%, while the MAPE between the predicted number and the ground truth boll weight was 8.83%. This approach provides a low-cost and efficient solution for generating high-quality plot-level cotton point clouds, offering a valuable tool for field-based cotton phenotyping and precision management.

Keywords. 3D reconstruction; 3D segmentation; NeRF; SAM; Plot-level cotton crop

1 Introduction

Cotton is the world's most important renewable fiber crop, with a global production value exceeding \$50 billion, supporting the livelihoods of 150 million cotton workers across 75 countries (Devoto et al., 2024; Kun et al., 2025). As the global population continues to grow, developing effective strategies to improve cotton yield has become a critical challenge in modern agriculture. Plant phenotypic information plays a key role in supporting breeding programs to increase cotton yield by providing essential data (Conaty et al., 2022; Voss-Fels et al., 2019). Currently, phenotypic data collection in cotton is primarily conducted through manual field sampling, which is time-consuming, labor-intensive, and prone to human error, especially under high-temperature outdoor conditions (Chen et al., 2025). Therefore, there is an urgent need for efficient methods to analyze cotton phenotypic traits.

Deep learning methods for 2D images analysis have been successfully applied in fruit recognition tasks. For instance, YOLOv5x achieved an average precision of 99.5% in kiwifruit detection (Gao et al., 2024). while YOLOv8n attained an average accuracy of 91.93% for strawberry detection (An et al., 2024). The spatial distribution of these fruits is relatively simple, which facilitates accurate detection in two-dimensional (2D) images. In contrast, fruits with more complex three-dimensional (3D) spatial distributions—such as blueberries and apples—are more difficult to detect comprehensively due to frequent occlusions in 2D projections (Ni et al., 2021; Zheng et al., 2024). Similarly, although grape clusters can be identified in 2D images, detecting individual grapes within a cluster remains highly challenging because of their dense 3D arrangement (Yang et al., 2025). Cotton bolls also grow in a 3D spatial configuration, especially at the plot level, where severe occlusions in 2D images significantly compromise detection accuracy.

Compared to 2D images, point clouds offer a significant advantage by mitigating occlusion and enabling detailed 3D analysis of plant traits. Recent studies have employed point cloud instance segmentation to extract a variety of phenotypic traits, such as branch length and canopy volume in apple trees, and plant height and leaf angle in maize (Jiang et al., 2025a; Miao et al., 2021). Segmenting individual apples in point clouds facilitates accurate fruit counting and volume estimation (Dong et al., 2023). High-precision 3D point clouds are a prerequisite for successful segmentation. Terrestrial Light Detection and Ranging (LiDAR) systems can directly acquire high-quality point clouds; however, their high cost and slow acquisition speed hinder large-scale applications (Jiang et al., 2025b). In contrast, RGB-D cameras provide a more cost-effective alternative, but their limited resolution makes it difficult to capture fine-scale details such as branches (Bloch et al., 2025). Therefore, developing an efficient and low-cost 3D reconstruction method is critical for advancing high-throughput plant phenotyping and trait analysis.

Multi-view 3D reconstruction demonstrates great potential in balancing sensor cost and model resolution. This approach reconstructs the 3D structure of an object or scene by analyzing and processing 2D images captured from multiple overlapping viewpoints, typically through a classical combination of Structure from Motion (SfM) and Multi-View Stereo (MVS). SfM is primarily used to estimate camera parameters and generate a sparse point cloud, while MVS further refines the geometric details to produce a high-precision dense point cloud or mesh model. This reconstruction pipeline has been successfully applied to the 3D modeling of individual plants such as carex and kale (Yang et al., 2024). Several software tools that integrate SfM and MVS techniques—such as COLMAP (Schonberger and Frahm, 2016), PIX4D Mapper (Pix4D S.A., Lucerne, Switzerland), and Agisoft Metashape (Agisoft LLC, St. Petersburg, Russia)—have been widely adopted. By importing 2D images, these tools can automatically generate high-quality 3D models and have been applied to crops such as cotton and maize (Sun et al., 2020; Wu et al., 2020).

Neural Radiance Fields (NeRF) is a novel approach that leverages neural networks to implicitly represent 3D scenes (Mildenhall et al., 2020). By constructing a continuous volumetric radiance field, NeRF replaces traditional geometric techniques such as MVS, surface reconstruction, and texture mapping, thereby simplifying the pipeline while enabling photorealistic 3D rendering. NeRF has been successfully applied to the high-fidelity reconstruction of various fruits, including dragon fruit, grapes, lychees, and oranges (Hu et al., 2024). This method supports not only the 3D reconstruction of individual fruits but also entire plants. Moreover, its feasibility has been demonstrated at the plot level for crops such as corn and strawberries (Arshad et al., 2024; Zhang et al., 2024). However, the morphological complexity of cotton plants and the spatial distribution of cotton bolls pose greater challenges, necessitating further validation of NeRF's reconstruction performance at the plot scale.

After reconstructing the plant 3D model, segmentation is a key step for trait extraction. The segmentation of 3D plants can mainly be divided into two types of methods. One method involves exporting point clouds from a NeRF model and then using these point clouds to train a point cloud segmentation model. For example, PointNet++ (Qi et al., 2017) can be used

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to segment cotton bolls from cotton plants (Jiang et al., 2022). However, this approach requires extensive annotation of point clouds, leading to significant time spent on data labeling, particularly when labeling plants at the plot level, which is even more challenging. Additionally, the scarcity of data is also a notable issue. Another approach leverages multi-view 3D reconstruction techniques. It first obtains the target's 2D masks from 2D images, then maps these 2D masks into 3D space, subsequently segmenting the object in 3D (Jiang et al., 2024). This approach has already been applied to fruits such as apples and blueberries (Gené-Mola et al., 2020; Meyer et al., 2024; Ni et al., 2021). Considering the limited volume of our plot-level data and the challenges associated with 3D annotation, the second approach is more suitable for our data analysis. Combining NeRF technology to segment plot-level cotton data is promising, but it has not yet been explored by researchers.

To fill the knowledge gap, the overall goal of this study was to leverage NeRF for plot-level cotton plant 3D reconstruction and to develop a novel method for cotton boll segmentation by mapping 2D masks into 3D space. The specific objectives were to (1) design a data collection method based on ground robots equipped with multiple cameras and another method using a smartphone; (2) implement plot-level 3D reconstruction (NeRF) of cotton plants; (3) develop a 2D instance annotation strategy for cotton bolls based on SAM; (4) conduct plot-level cotton boll counting and evaluation.

2 Materials and Methods

2.1 Experimental field setup and data collection

The data used in this experiment were collected on October 25, 2024, in Tifton, Georgia, USA (Fig. 1a). The experimental field consisted of 160 plots (Fig. 1b), comprising 32 different genotypes, with each plot containing a single genotype. The center point of each plot was used as a reference, with a row spacing of 6 feet and a column spacing of 25 feet (Fig. 1e). Each plot measured approximately 18 feet in length. A defoliant was applied to the entire field one week prior to data collection. In some plots, most of the leaves had already fallen off (Fig. 1c), while others still retained a substantial amount of foliage (Fig. 1d). Nearly all bolls had already opened, with only a few remaining green. These immature bolls were excluded from yield measurements in this study. Data for testing were collected from the first, second, and seventh rows.

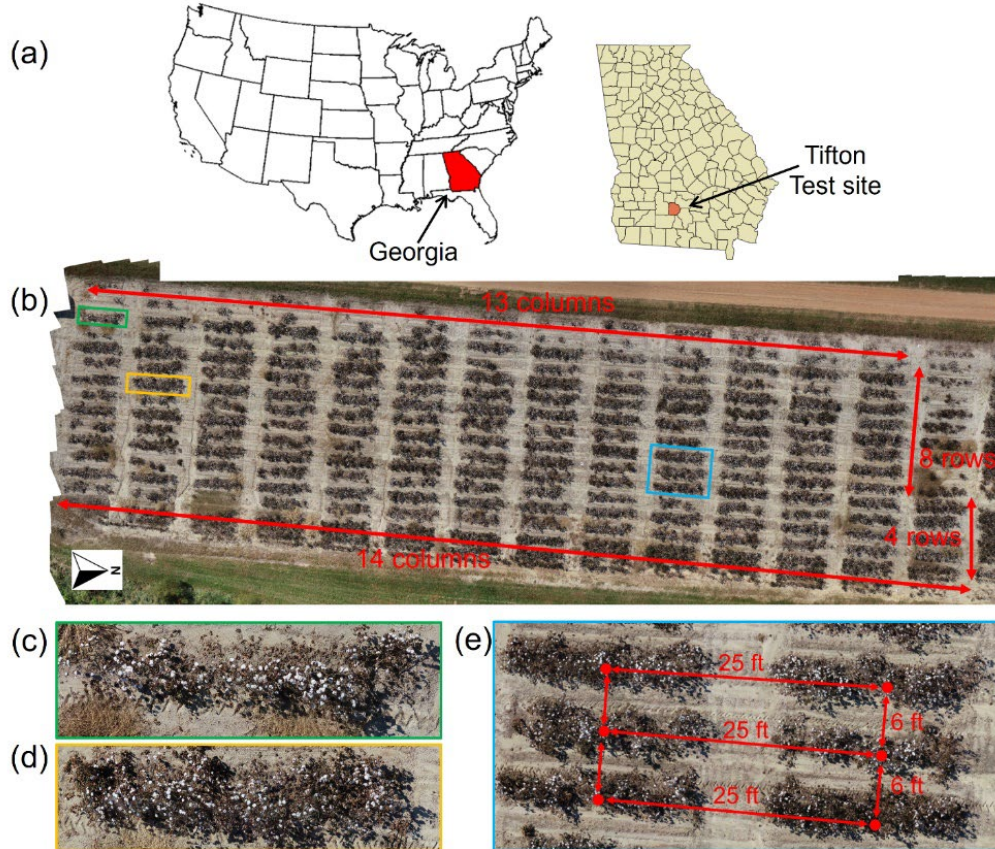


Fig. 1. Experimental field. (a) Data collection location. (b) The planting layout of plot-level cotton plants in the experimental field. (c) and (d) show two different plots. The leaves have completely fallen off, making the plot appear relatively sparse in (c). Some leaves remain on the plants, making the plot look denser in (d). (e) The red dots in the figure represent the center point of each plot. The distance between adjacent rows is 6 ft, and the distance between adjacent columns is 25 ft.

Two data collection methods were employed in this experiment. The first was a high-throughput approach utilizing a ground robot equipped with multiple cameras, while the second was a low-cost method based on a smartphone. The ground robot, MARS (Xu and Li, 2022), was equipped with ten cameras, which were mounted approximately evenly around the

canopy bracket. Due to the limited availability of cameras, two different types were used in this setup. The cameras marked with red circles in Fig. 2a indicate five Fuji X-A10 cameras (Fujifilm Holdings Corporation, Tokyo, Japan), whereas the blue circles represent five Lumix DMC-G7 cameras (Panasonic Corporation, Osaka, Japan). All cameras recorded videos at 30 frames per second for plant scanning. The Fuji cameras captured video at a resolution of 1920×1080 , while the Lumix cameras recorded at 3840×2160 . Each camera was adjusted to maximize its field of view, and the angles were fine-tuned to ensure that the plants were centered in the image frame (Fig. 2c). On average, it took approximately 40 seconds to scan a single plot using this method.

Video collection using a smartphone provides greater flexibility. The smartphone was mounted on a handheld stabilizer and used to scan the plants directly. Videos were recorded at a resolution of 3840×2160 and a frame rate of 30 frames per second. The field of view was adjusted to ensure that the entire scanning area was included in the image. For instance, when capturing side views of the plant, both the top and bottom of the cotton plant were visible within the frame (Fig. 2b). For top views, the plant was positioned as close to the center of the image as possible. The field of view was kept consistent across all individual plots. On average, it took approximately 130 seconds to scan a single plot using this method.

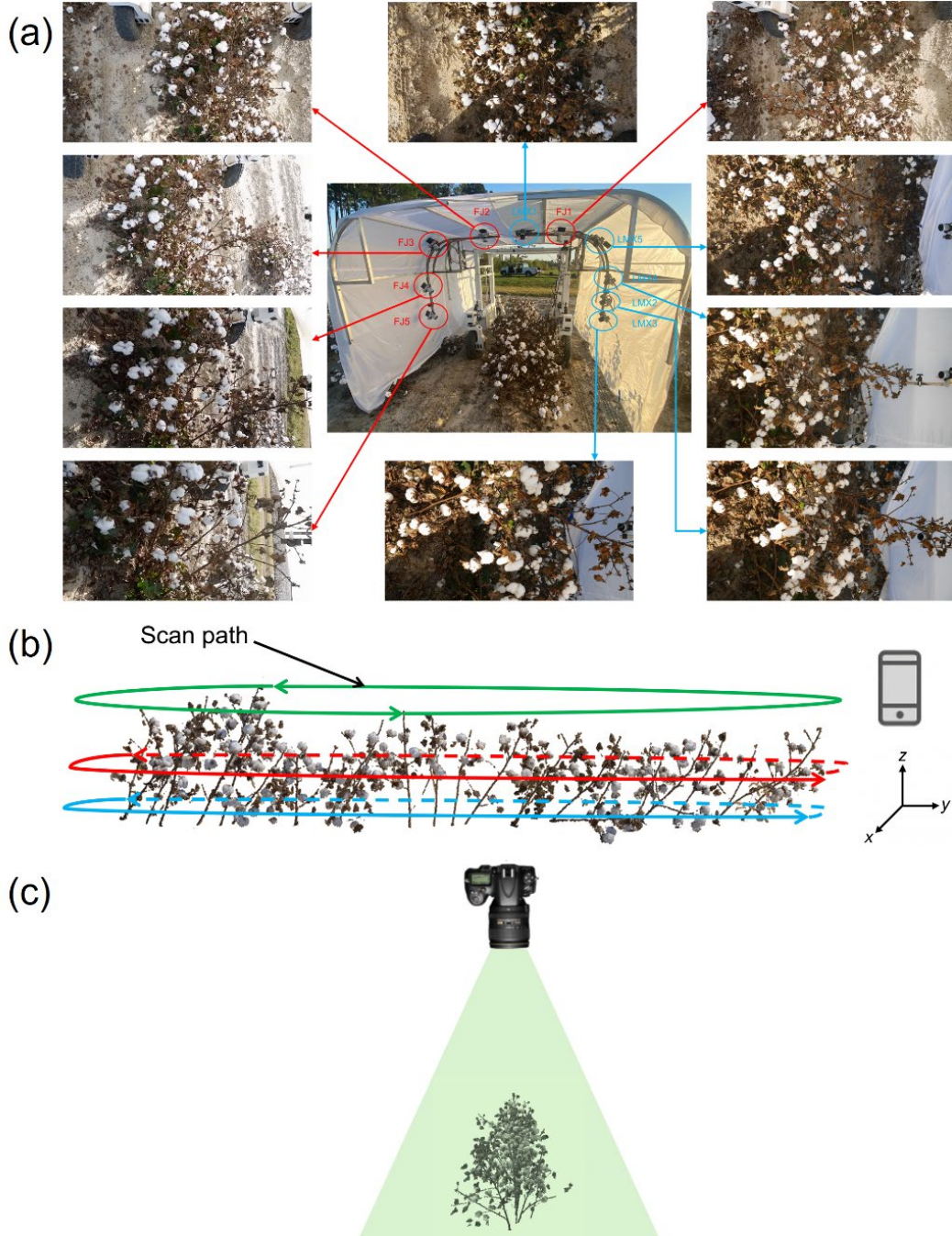


Fig. 2. Multi-view image acquisition system. (a) The MARS robot is equipped with 10 cameras, with Fuji cameras marked by red circles and Lumix cameras marked by blue circles. (b) Data collection path using a smartphone. The green path represents the top view, while the red and blue lines indicate side views at different heights. Data were collected at three height levels: upper, middle, and lower. (c) Data acquisition guidelines. The plant should remain within the camera's field of view and be positioned at the center of the frame whenever possible.

2.2 Plot-level cotton plant 3D reconstruction based on NeRF

Neural Radiance Fields (NeRF) is a novel implicit method for 3D reconstruction that models the appearance of a scene from any spatial position and viewing direction using a trained neural network (Mildenhall et al., 2020). The input to NeRF consists of a 3D coordinate (x, y, z) and a viewing direction (θ, ϕ) , while the output is the color (R, G, B) and volumetric density (σ) at that point in the specified direction. By sampling points along camera rays and applying volume rendering techniques, NeRF can synthesize photorealistic images from arbitrary viewpoints using multiple 2D images, while preserving detailed geometric and texture information.

NeRF was employed to reconstruct cotton plants at the plot level and to generate corresponding point clouds. Images were extracted from recorded videos and standardized to a resolution of 1920×1080 . For multi-camera recordings, approximately 90 frames were uniformly sampled from each video, ensuring over 90% overlap between adjacent frames. In the case of smartphone-captured videos, 500 frames were evenly extracted across the video duration, with similarly high overlap ($>90\%$) between neighboring frames. The extracted multi-view images were processed using COLMAP, which performed feature extraction, feature matching, and sparse reconstruction to estimate both intrinsic and extrinsic camera parameters, ultimately generating a sparse point cloud of the entire scene. Three-dimensional (3D) reconstruction and evaluation of plot-level cotton plants were conducted using Nerfstudio (version 1.1.4; <https://docs.nerf.studio/>). Based on the estimated camera parameters and RGB images, the Nerfacto (Tancik et al., 2023) model was utilized to learn a 3D representation of the scene. Each scene was trained for 30,000 iterations to complete the reconstruction process.

2.3 Plot-level 3D cotton boll segmentation and counting

To enable cotton boll counting at the plot level, this study proposed a workflow based on 2D-to-3D mapping, effectively addressing the challenge of limited 3D annotation data. First, the YOLOv11x model was used to predict 2D masks of cotton bolls from RGB images. Then, FruitNeRF was trained using the RGB images and their corresponding binary cotton boll masks to segment the cotton boll point cloud from the 3D plot-level model. Finally, Hierarchical Density-Based Spatial Clustering of Applications with Noise (HDBSCAN) was applied to the point cloud for clustering and boll counting.

2.3.1 A cotton boll 2D mask annotation method based on a foundation model

Building on the laboratory's previous research (Tan et al., 2024), this study developed a semi-automatic 2D mask annotation method for cotton bolls based on the Segment Anything Model (SAM; Kirillov et al., 2023). First, a YOLOv11x object detection model was trained using a cotton boll detection dataset from previous research. These existing images were collected in 2022 using a top-view acquisition method. They were already annotated, with each cotton boll labeled with a 2D bounding box. Second, the trained model was used to predict cotton bolls on unlabeled new data, where each detected boll was assigned a 2D bounding box (Fig. 3).

Next, the predicted 2D bounding boxes were provided as prompts to SAM to generate 2D masks of the cotton bolls. SAM is a foundation model with zero-shot learning capabilities. Unlike supervised learning models that rely on large amounts of pixel-level annotations, SAM follows a prompt-based segmentation paradigm, where user-provided prompts—such as points, bounding boxes, or text—guide the segmentation process. Trained on a large-scale dataset containing over 11 million images and more than 1 billion masks, SAM demonstrates strong generalization to unseen tasks and objects. This extensive training enables SAM to perform accurate segmentation without requiring task-specific fine-tuning. For the new data collected in 2024, the 2D bounding boxes predicted by YOLOv11x were used as prompts, while for the 2022 data, manually annotated 2D bounding boxes were provided as prompts to SAM to generate 2D masks of cotton bolls.

The 2D masks generated by SAM required further manual refinement. When cotton bolls were occluded by the main stem or branches, the generated masks were often incomplete and needed manual correction. YOLOv11x occasionally produced incorrect detections or missed cotton bolls entirely, leading to inaccurate or missing masks. To address these issues, the SAM-generated masks and corresponding RGB images were imported into Roboflow (<https://roboflow.com/>) for manual refinement. Finally, a 2D segmentation dataset of cotton bolls was obtained.

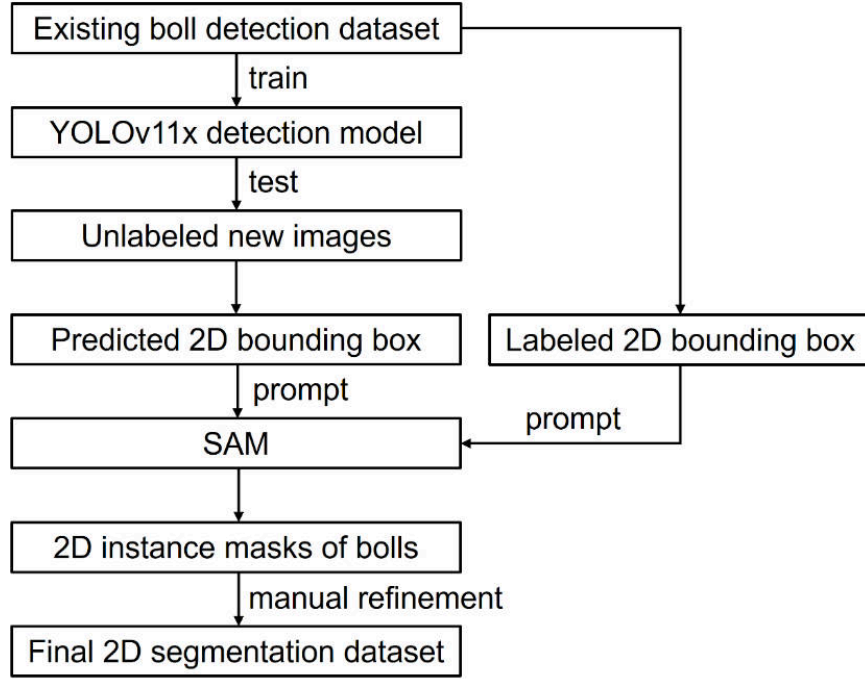


Fig. 3. Semi-automatic annotation pipeline for a 2D segmentation dataset using SAM.

The 2D segmentation dataset of cotton bolls was used to train the YOLOv11x segmentation model, which was then used to segment 2D masks of cotton bolls in the predicted images. Table 1 presents a summary of the cotton boll segmentation dataset. In this study, the dataset was divided into a training set and a validation set. The YOLOv11x model was first trained on the 2022 dataset and then fine-tuned on the 2024 dataset to obtain the final cotton boll segmentation model.

Table 1. Summary of cotton boll segmentation dataset for training and validation.

	Training images	Validation images
2022 data	606	260
2024 data	98	27

2.3.2 Point cloud semantic segmentation based on FruitNeRF

FruitNeRF (Meyer et al., 2024) is a unified fruit segment framework that leverages NeRF to construct a 3D representation of the scene and incorporates semantic information for exporting the fruit point cloud. The model first takes an unordered set of RGB images and generates semantic segmentation masks for the fruits in each image. In the workflow of this study, YOLOv11x was used to obtain 2D masks of cotton bolls. The RGB images and the corresponding binary masks of the cotton bolls were combined to train FruitNeRF, enabling the entire scene to be enriched with semantic information of the fruits (Fig. 4). The trained FruitNeRF was then uniformly sampled to generate the cotton boll point cloud.

FruitNeRF extends the original NeRF framework by introducing a semantic field, in addition to the existing density and appearance fields. This semantic field encodes fruit-related information within the 3D scene, allowing each spatial location to be identified and labeled as either a cotton boll or background. It is implemented using a multi-layer perception (MLP) that approximates the semantic logits $S(r)$ or each pixel based on its 3D position. To estimate the semantic logits $S(r)$ for a pixel, volume density and semantic data are integrated over K sample points along the ray r using numerical methods.

$$S(r) = \sum_{k=1}^K T(t_k) \alpha(\sigma(t_k) \delta_k) s(t_k) \quad (1)$$

$T(t_k)$ represents the accumulated transmittance from t_n to t_k . $\delta_k = t_{k+1} - t_k$ represents the distance between adjacent sampled points. $\sigma(t_k)$ represents the volume density at the point t_k . $\alpha(\sigma(t_k) \delta_k)$ represents the light absorption value at the sampled point, calculated as: $\alpha(x) = 1 - \exp(-x)$. $s(t_k)$ represents the semantic value at the sampled point t_k , which is obtained through the MLP network and represents the semantic information of that location.

The cotton boll semantic field consists of 3 layers, with a hidden layer size of 128 neurons. The input layer has a dimension of 30, derived from a 30-dimensional latent vector from the density field, and the output dimension of the neural network is 64. This output serves as input for the neural segmentation head, which further reduces the dimension to a single class. When exporting the cotton boll point cloud, the number of rays to be sampled per batch during the point cloud generation process is set to 3000, and the number of points to be sampled along each edge of the 3D scene (bounding box) is set to 6000. Afterward, the cotton boll point cloud for a single plot is manually cropped to include only the cotton boll points of the plot of interest.

2.3.3 Clustering cotton boll point clouds for counting

The cotton boll points for each plot were obtained in the previous step. To achieve boll counting, instance segmentation was performed using a clustering method. Hierarchical Density-Based Spatial Clustering of Applications with Noise (HDBSCAN) is a density-based clustering algorithm that effectively identifies clusters of arbitrary shapes and varying densities, while also handling noise points (McInnes et al., 2017). Combining the advantages of DBSCAN and hierarchical clustering, HDBSCAN eliminates the need for predefined cluster numbers or distance thresholds. Its reachability distance design enables it to handle clusters with varying densities and automatically determines the number of clusters by constructing a minimum spanning tree and performing hierarchical clustering. This makes HDBSCAN particularly suitable for datasets with clusters of varying densities, especially in high-dimensional, non-uniformly distributed point cloud data. It is ideal for 3D segmentation and counting tasks in complex plant structures with an unknown number of targets. In this study, the `min_cluster_size` and `min_samples` parameters of HDBSCAN were set to 20 and 5, respectively, based on empirical values derived from experimentation.

2.4 Implementation details

The training of the point cloud deep learning model was conducted on the HiPerGator high-performance computing cluster, which was equipped with 8 AMD EPYC Rome CPU cores and a single NVIDIA DGX A100 GPU node (80 GB), running on a Linux operating system. The YOLOv11x segmentation model was trained on the same cluster. The software environment included Python 3.10, PyTorch 1.12.1, and CUDA 11.3. The training parameters were as follows: the input image size was 1280×1280, the batch size was 8, and the total number of training epochs was 300. All other parameters were set to their default values. The 3D reconstruction of cotton plants using the NeRF model was tested on the Nerfstudio platform, with FruitNeRF tested on the official code provided.

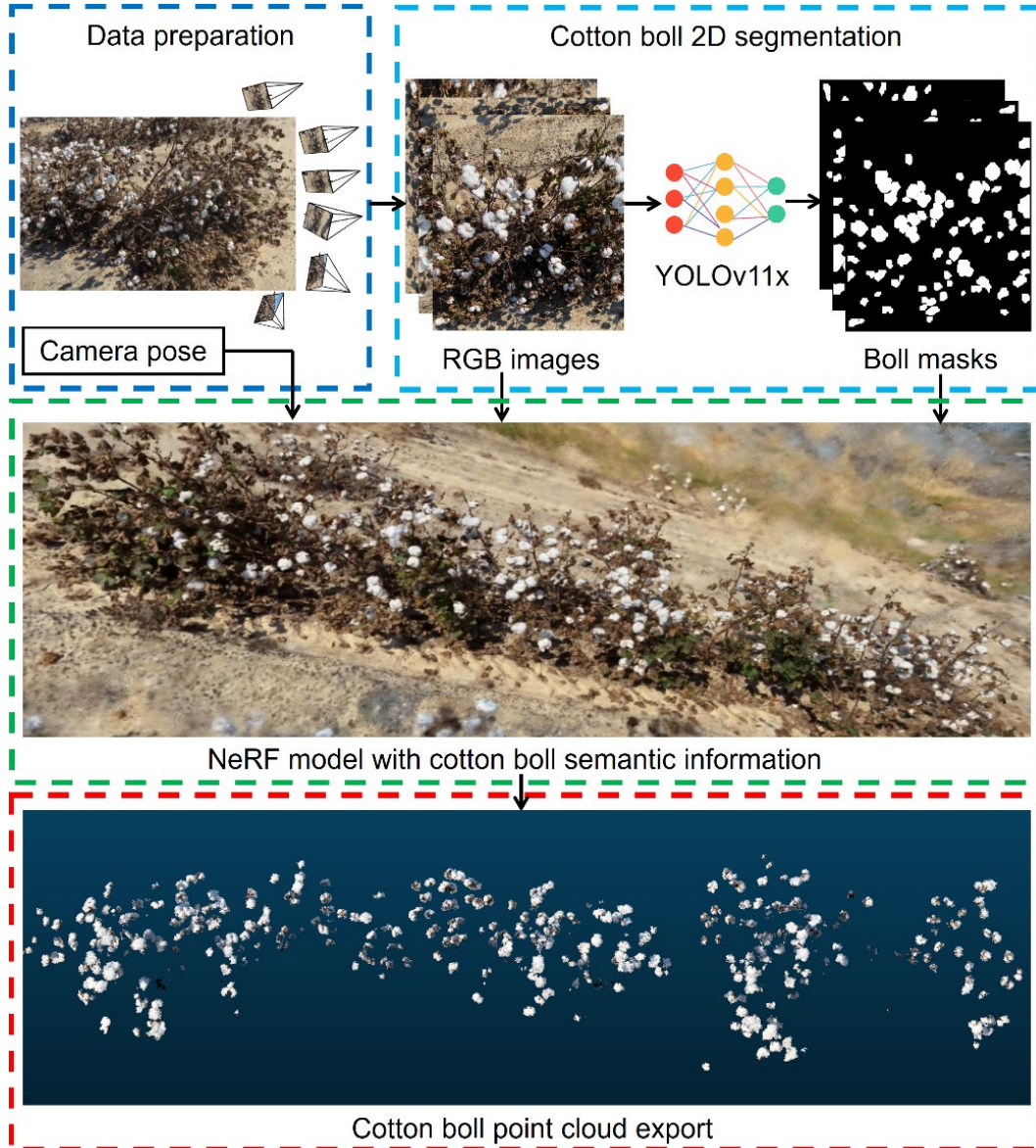


Fig. 4. Plot-level cotton boll point cloud segmentation workflow. First, camera parameters were estimated by Structure-from-Motion (SfM) using multi-view images. Then, YOLOv11x was employed to extract semantic masks of cotton bolls. Next, the posed RGB images and the corresponding semantic masks were used to train a Neural Radiance Field (NeRF). Finally, a dense cotton boll point cloud was generated by uniformly sampling the NeRF model with embedded cotton boll semantic information.

2.5 Evaluation Metrics

The Mean Squared Error (MSE) was adopted to quantify the difference between two images. A lower MSE value indicates a higher level of similarity. In this context, m and n denote the height and width of the image, respectively. $I(i, j)$ represents the pixel intensity at position (i, j) in the original image, while $K(i, j)$ corresponds to the pixel intensity at the same position in the processed image. The Peak Signal-to-Noise Ratio (PSNR), a widely used metric for assessing image quality, reflects how closely the processed image matches the reference image. A higher PSNR value signifies better image quality (Eq. 3). PSNR is computed based on the MSE, where MAX stands for the maximum possible pixel value in image I .

$$MSE = \frac{1}{mn} \sum_{i=0}^{m-1} \sum_{j=0}^{n-1} [I(i, j) - K(i, j)]^2 \quad (2)$$

$$PSNR = 10 \cdot \log_{10} \left(\frac{MAX_I^2}{MSE} \right) \quad (3)$$

The Structural Similarity Index (SSIM) was employed to evaluate image similarity in terms of brightness, contrast, and structural information. In this metric, μ_x and μ_y represent the mean pixel intensities of images x and y , respectively, while σ_x and σ_y denote their variances. σ_{xy} refers to the covariance between the two images. The constants c_1 , c_2 , and c_3 are used to stabilize the calculations of brightness, contrast, and structure components. In addition, the Learned Perceptual Image Patch Similarity (LPIPS), also known as perceptual loss (Zhang et al., 2018), compares the deep features of image patches, aiming to reflect perceptual similarity as captured by the human visual system.

$$SSIM(x, y) = \frac{(2\mu_x\mu_y + c_1)(2\sigma_{xy} + c_2)(\sigma_x\sigma_y + c_3)}{(\mu_x^2 + \mu_y^2 + c_1)(\sigma_x^2 + \sigma_y^2 + c_2)(\sigma_x\sigma_y + c_3)} \quad (4)$$

The accuracy of cotton boll count estimation was evaluated using root mean square error (RMSE), mean absolute error (MAE), and mean absolute percentage error (MAPE). In these formulas, N_i denotes the predicted count for the i -th sample, m_i denotes the corresponding ground truth, and n represents the total number of evaluated samples.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (N_i - m_i)^2} \quad (5)$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |N_i - m_i| \quad (6)$$

$$MAPE = \frac{100\%}{n} \sum_{i=1}^n \left| \frac{N_i - m_i}{m_i} \right| \quad (7)$$

3 Results and Discussion

3.1 3D reconstruction result evaluation

Various types and combinations of cameras can be utilized for 3D reconstruction, each potentially yielding different reconstruction qualities. In the present study, due to constraints on the number of available devices, multiple types of cameras were employed. Consequently, it was necessary to compare the reconstruction performance across different camera configurations. Table 2 summarizes the evaluation results of the NeRF model applied to data from different cameras. Given the inherently complex and dynamic nature of field environments, the reconstruction quality exhibited variability among different plots. To account for this, ten representative plots were selected, and the average performance metrics were calculated. Although the reported metrics are primarily designed to assess the rendering quality of NeRF, the point clouds were directly derived from the trained NeRF models. Therefore, these metrics can serve as indirect indicators of the geometric quality of the exported point clouds.

In the overall evaluation of NeRF reconstruction, the 3D results based on smartphone data and those using five Lumix cameras outperformed the reconstructions generated using five Fuji cameras and the combination of ten cameras. The use of different types of cameras can lead to poor reconstruction quality due to color discrepancies between camera sensors. It is not recommended to use multiple cameras of different types for NeRF-based reconstruction. This is primarily due to inherent differences in camera sensor characteristics, which can result in variations in color representation across captured images. Such discrepancies are reflected in the reconstructed point clouds, as illustrated in Fig. 5. Specifically, the point

cloud generated using five Lumix cameras appears darker in color (Fig. 5c), whereas the one reconstructed with five Fuji cameras is noticeably brighter (Fig. 5d). When both camera types are used simultaneously, the resulting point cloud displays inconsistent and visually chaotic color patterns (Fig. 5b). These findings underscore the importance of maintaining sensor consistency in multi-camera data acquisition setups for improved reconstruction quality.

In the quantitative evaluation results, the NeRF reconstruction quality of the data collected by five Lumix cameras showed a PSNR 1.0 higher, an SSIM 0.14 higher, and an LPIPS 0.05 lower compared to the smartphone data. The comparison of the visualization results in Fig. 5 reveals that both data collection methods achieved good 3D reconstruction results. Since the smartphone scanned the top view and both side views of the entire plot, while the five Lumix cameras only scanned the top view and a single side view, this experiment first tested the cotton boll counting results using the smartphone-collected data, and future tests will focus on the cotton boll counting results using the Lumix camera-collected data.

Table 2. Comparison of NeRF model results reconstructed from images captured by different cameras.

	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
10 cameras	14.5190	0.3766	0.6915
5 Lumix cameras	17.2100	0.4883	0.5435
5 Fuji cameras	14.6100	0.3186	0.6314
Smartphone	16.2105	0.3435	0.4909

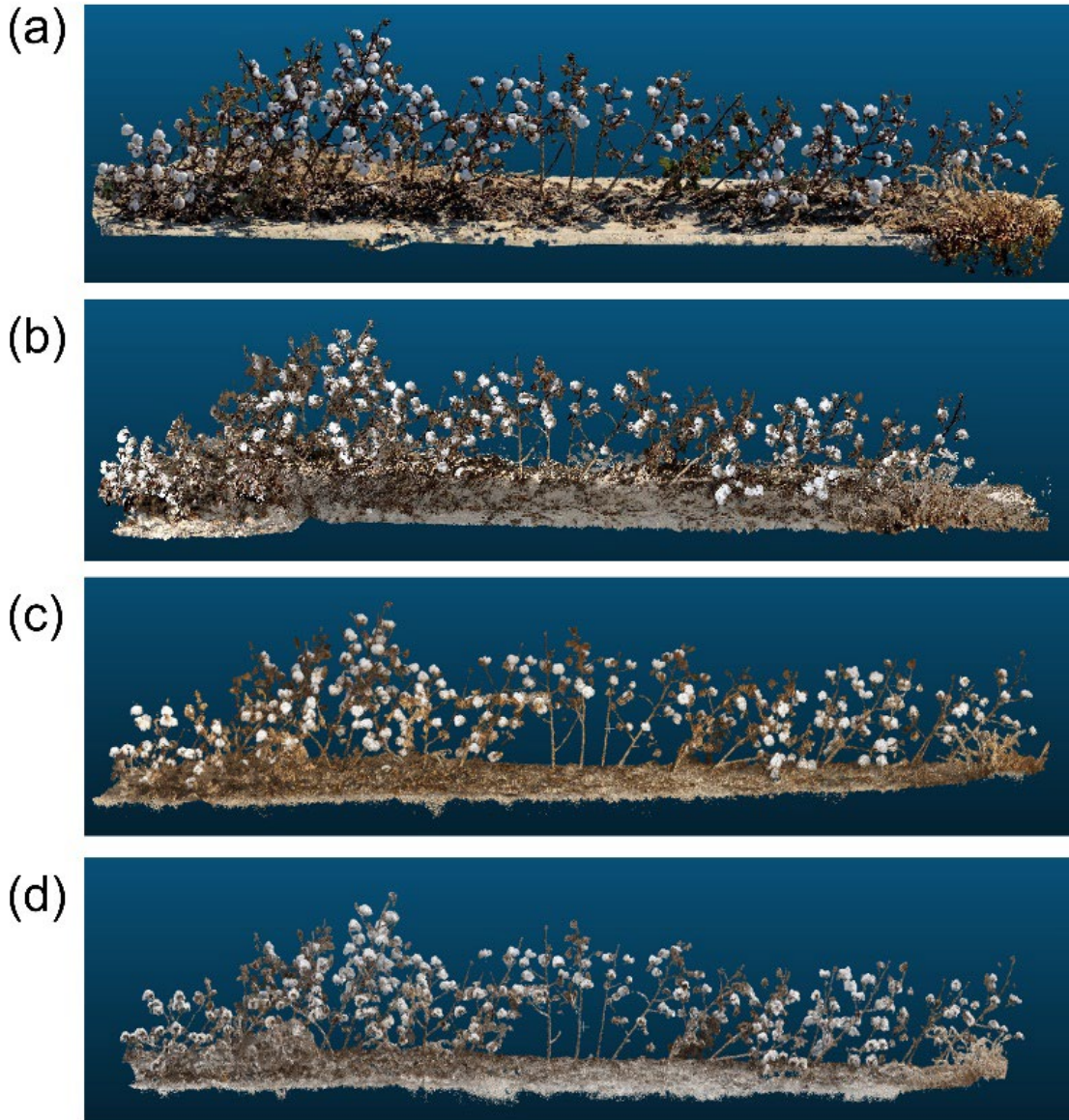


Fig. 5. Plot-level cotton plant point clouds reconstructed using different camera combinations. (a) Single smartphone. (b) Ten cameras of two different types. (c) Five Lumix cameras. (d) Five Fuji cameras.

3.2 2D instance segmentation evaluation

The 2D segmentation approach based on SAM and YOLOv11x developed in this study substantially accelerated the data annotation process and reduced the annotation burden for new datasets. Compared with conventional manual labeling methods, this approach significantly decreased the labor required. According to our rough estimates during the annotation process, the annotation time per image was reduced by at least 50%. Furthermore, the proposed workflow is highly adaptable and can be easily extended to other crop species or datasets, demonstrating strong flexibility and scalability.

Fine-tuning the YOLOv11x segmentation model with new data from 2024 led to a substantial performance improvement over the model trained solely on the 2022 dataset. When evaluated on the 2024 dataset, the model trained exclusively on 2022 data achieved an mAP50 of only 72.8%. After fine-tuning with the 2024 data, the mAP50 increased to 84.5%, representing a notable improvement of 11.7% (Table 3). As illustrated by the segmentation results of smartphone-captured images in Fig. 6, cotton bolls were accurately segmented in both top-view (Fig. 6a, c) and side-view (Fig. 6b, d) images. The results also indicate that cotton bolls from adjacent plots were occasionally segmented. However, these instances can be manually filtered out during the subsequent 3D mapping process.

Table 3. 2D instance segmentation with YOLOv11x.				
	P	R	mAP50	mAP50-95
2022 data	0.732	0.661	0.728	0.428
2024 data	0.812	0.778	0.845	0.548

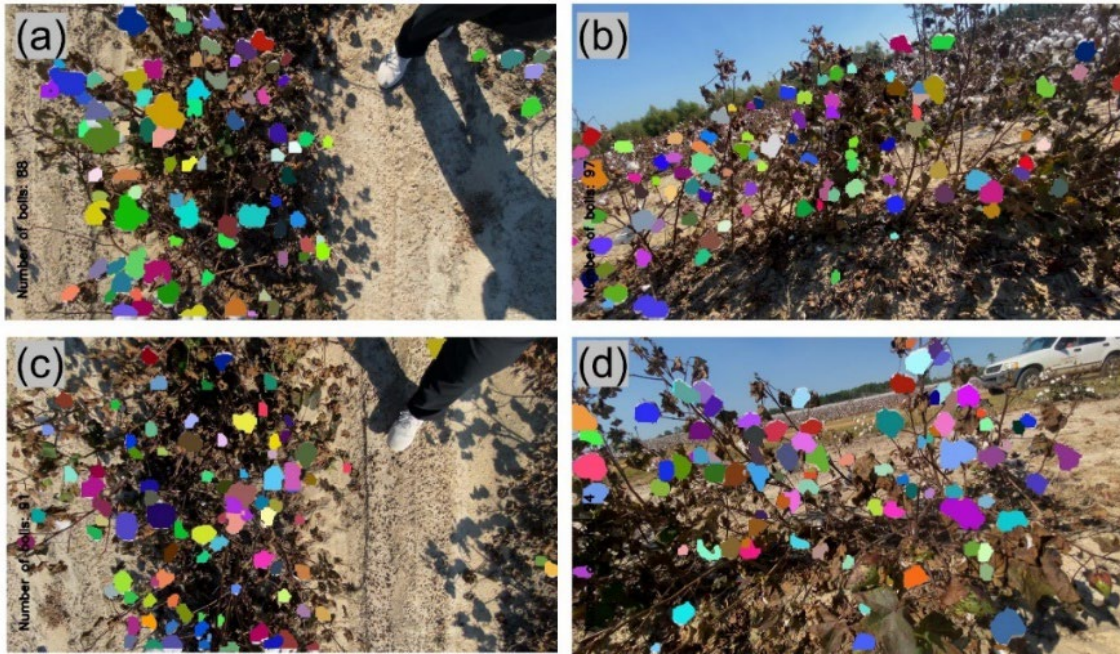


Fig. 6. YOLOv11x cotton boll 2D instance segmentation visualization. (a) and (c) are top views, while (b) and (d) are side views.

3.3 3D instance segmentation and boll counting using smartphone data

FruitNeRF enables the segmentation of cotton boll point clouds at the plot level, effectively addressing both occlusion issues present in 2D imagery and the scarcity of annotated 3D data. By applying HDBSCAN clustering to the cotton boll point clouds segmented by FruitNeRF, instance-level segmentation and accurate counting of cotton bolls were achieved. Figure 7 presents segmentation results for two representative plots. A key distinction between the two plots is that one has completely shed its leaves (Fig. 7a–c), while the other retains a substantial number of leaves (Fig. 7d–f). Despite this difference in foliage, accurate cotton boll counts were obtained in both cases. Figures 7a and 7d display the HDBSCAN clustering results applied to the point clouds exported by FruitNeRF, where each identified cluster is assigned a unique color and enclosed within a 3D bounding box. Figures 7b and 7e project the clustering results back onto the original plot-level point clouds, showing the side view of the plants. Figures 7c and 7f provide the corresponding top views, further illustrating the spatial distribution of the segmented cotton bolls.

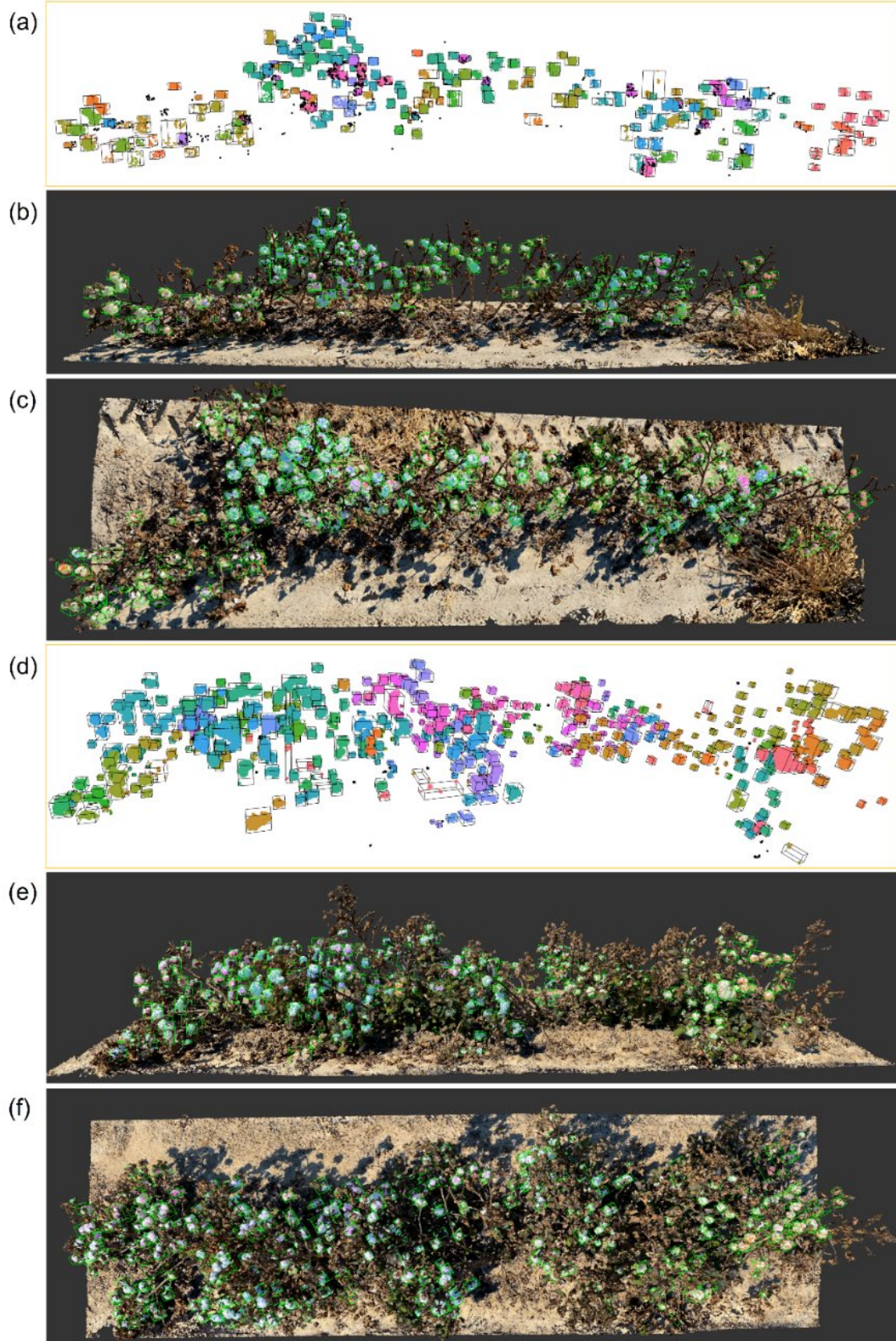


Fig. 7. Cotton boll segmentation and clustering results based on data collected using a smartphone. (a)-(c) and (d)-(f) represent two different plots. Within each plot case, the three sub-figures show the clustering results of cotton bolls, the side view of the plot, and the top view of the plot, respectively.

The accuracy of cotton boll counting is not only related to the segmentation precision of FruitNeRF, but also depends on the clustering results of HDBSCAN. Fig. 8 illustrates a common issue in the clustering step of the cotton boll counting pipeline. As highlighted by the yellow circle, two adjacent cotton bolls are incorrectly grouped into a single cluster by

HDBSCAN. This occurs due to the close spatial proximity and partial occlusion between the bolls, which makes it difficult for the density-based algorithm to distinguish them as separate instances. Such cases can lead to slight underestimation in the final boll count and suggest the need for further refinement in clustering parameters or post-processing strategies.

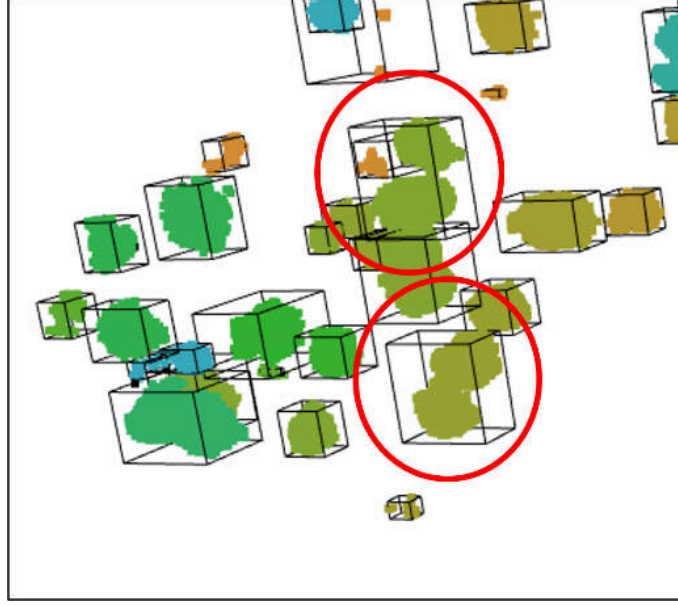


Fig. 8. Limitations of cotton boll clustering results. The two cotton bolls within the red circle were grouped into one cluster.

The workflow developed in this study enables accurate estimation of cotton yield (Fig. 9). In the evaluation conducted on 10 plots, the predicted number of cotton bolls showed a strong correlation with manual counts ($R^2 = 0.94$), achieving a high accuracy with a mean absolute percentage error (MAPE) of 4.41%. The predicted boll numbers were further used to estimate the yield for each plot. A regression analysis between the predicted boll numbers and the actual measured weights revealed a strong correlation ($R^2 = 0.92$), with a prediction error of 8.83%. The average weight estimation error per plot was less than 130 grams.

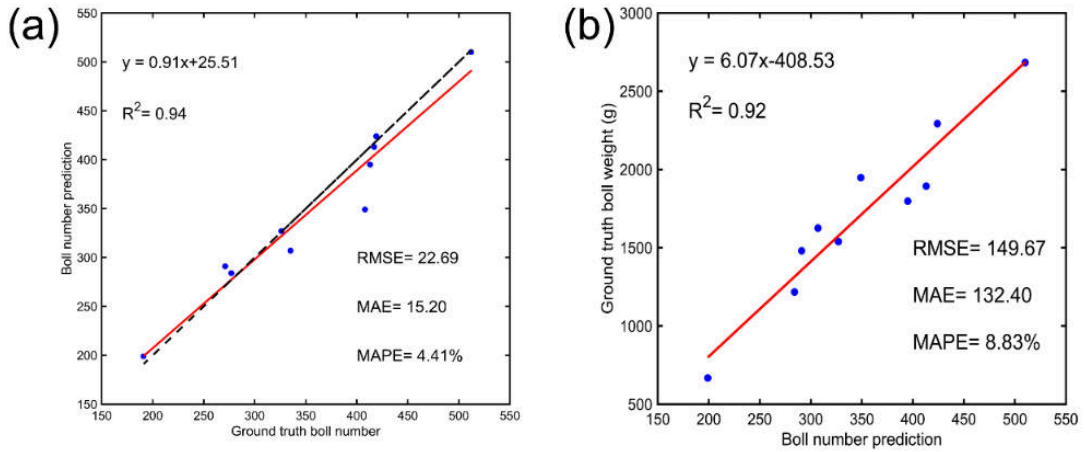


Fig. 9. Plot-level cotton yield estimation. (a) Linear regression between the predicted and ground truth number of cotton bolls. (b) Linear regression between the predicted number and the ground truth weight of cotton bolls. The red solid line represents the fitted line, the black dashed line indicates the 1:1 line, and the blue dots represent different plots.

4 Conclusion

This study proposes a novel plot-level workflow for 3D reconstruction and cotton boll counting. The workflow enables high-fidelity NeRF reconstruction of cotton plants and exports a complete plot-level point cloud of the cotton crop. By obtaining high-quality 2D masks of cotton bolls from RGB images and projecting them onto the 3D point cloud using FruitNeRF, the method effectively bridges 2D segmentation with 3D analysis to achieve plot-level boll counting and yield estimation. This innovative approach provides a new tool for 3D plant phenotyping and high-resolution plant mapping, which can be extended to other crops. In future work, we will test more plot-level data to evaluate the robustness of the cotton boll counting method.

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